

Hirad Daneshvar

Applied ML Researcher | Trustworthy AI

Ottawa, Ontario, Canada | hirad.daneshvar@inrs.ca

<https://linkedin.com/in/hiraddaneshvar/> | <https://github.com/hi-rad> | <https://hdaneshvar.com>

Detailed CV: https://hdaneshvar.com/Hirad_Daneshvar_CV.pdf

SUMMARY

Ph.D. in Machine Learning/Computer Engineering and current Postdoctoral Fellow with **5+ years of full-stack software engineering** experience. Specialized in Graph Neural Networks (GNNs), Uncertainty Quantification, and LLM reliability. Proven ability to bridge the gap between state-of-the-art AI research and production-ready, Dockerized backend architectures.

WORK EXPERIENCE

Postdoctoral Research Fellow

Sep. 2025 – Present

INRS-UQO UMR Research Lab

Gatineau, QC, CA

- Member of the Multisensory Signal Analysis and Enhancement (MuSAE)
- Doing research and development of **Safety-Critical Alignment** and **Uncertainty Quantification** for LLMs to ensure reliability in high-stakes interactions

Research Assistant

Jan. 2021 – Sep. 2025

Toronto Metropolitan University

Toronto, ON, CA

- Engineered novel **Trustworthy Graph Neural Networks** for complex system predictive modelling
- Designed and developed the **Graph Confidence Error (GCE)** framework to calibrate models, resulting in a 7-12% reduction in calibration error
- Designed and developed an efficient self-distillation-based **Uncertainty Quantification** method, resulting in a 75-90% test-time efficiency
- Led a project on **trustworthy LLM ChatBots** for Mental Health Triaging

Software Developer

Jul. 2020 – Mar. 2021

Payrad Smart Solutions

Tehran, IR

- Developed databases using **MySQL** in a FinTech-related startup
- Developed and maintained **Dockerized** web applications back-end using **PHP/NodeJS** and the **Laravel/NestJS** Framework to help speed up wage payments, serving thousands of companies

Software Developer & Team Lead

Sep. 2018 – Jul. 2020

Fanap Plus

Tehran, IR

- Developed databases using **MySQL** and **MongoDB**
- Developed and maintained **Dockerized** web applications back-end using **PHP** and the **Laravel** Framework
- Developed and maintained phone top-up application and soccer league news and prediction application serving millions of users

Software Developer

Jul. 2013 – Apr. 2018

Mandegar System

Tehran, IR

- Designed and developed an e-commerce system with features such as dynamic product fields using **PHP**, **MySQL**, and **MongoDB**

EDUCATION

Toronto Metropolitan University

Toronto, ON, CA

Ph.D. in Machine Learning / Computer Engineering

Jan. 2021 – Oct. 2025

Supervisor: Dr. Reza Samavi (Trustworthy AI Lab)

Dissertation: Trustworthy Graph Neural Networks for Healthcare Predictive Modelling

Islamic Azad University - Central Tehran Branch

Tehran, IR

M.Sc. in Computer Engineering

Sep. 2016 – Sep. 2020

Supervisor: Dr. Reza Ravanmehr

Thesis: A Social Hybrid Recommendation System using LSTM and CNN

Islamic Azad University - Central Tehran Branch

Tehran, IR

B.S. in Computer Engineering

Sep. 2010 – Jan. 2016

SKILLS

Languages: Python, Java, PHP, JavaScript, HTML/CSS (Working Knowledge)

Databases: SQL, NoSQL, MongoDB, MySQL

Frameworks: PyTorch, FastAPI, PyG, Laravel, Yii

Core Machine Learning: Machine Learning, Deep Learning, Applied Research, Model Calibration, Uncertainty Quantification

Specialized Domains: Graph Neural Networks (GNNs), Large Language Models (LLMs), Recommendation Systems

Developer Tools: Git, Docker, VS Code, PyCharm

Libraries: Scikit-Learn, Pandas, NumPy, Matplotlib

ADDITIONAL INFORMATION

Reviewer: IEEE Transactions on Human-Machine Systems, Wiley Computational Intelligence, Second Symposium on Human Partnership with Medical AI (AAAI Summer Symposium Series 2023), Canadian AI 2025 & 2026, AliH 2026, IEEE IRI 2026

Teaching:

- Contract Lecturer for the Software Design and Architecture Course at Toronto Metropolitan University
- Contract Lecturer for the Software Systems Course at Toronto Metropolitan University
- Contract Instructor for the Web and Mobile Software Development Course at Carleton University

PUBLICATIONS

[C5] Czobit, C., **Daneshvar, H.**, Samavi, R., Doyle, T. E., Duncan, L., Pires, P., & Sassi, R. (2026). *Representation Effects in Child and Youth Mental Health Emergency Readmission Predictions*. Proceedings of the 39th Canadian Conference on Artificial Intelligence. (*Status: Accepted - In Press*)

[C4] **Daneshvar, H.**, & Samavi, R. (2025). *Efficient Subsampling for GNN Downstream Tasks*. Proceedings of the 17th Asian Conference on Machine Learning. PMLR, 304:1070-1085 [\[Link\]](#)

[C3] **Daneshvar, H.**, & Samavi, R. (2025). GNN's Uncertainty Quantification Using Self-distillation. In International Conference on AI in Healthcare (pp. 31-45). Cham: Springer Nature Switzerland. [\[Link\]](#)

[C2] **Daneshvar, H.**, & Samavi, R. (2024). *GCE: Confidence Calibration Error for Improved Trustworthiness of Graph Neural Networks*. Proceedings of the Canadian Conference on Artificial Intelligence. [\[Link\]](#)

[BC1] **Daneshvar, H.**, Boursalie, O., Samavi, R., Doyle, T. E., Duncan, L., Pires, P., & Sassi, R. (2024). *SOK: Application of machine learning models in child and youth mental health decision-making*. Artificial Intelligence for Medicine, 113-132. [\[Link\]](#)

[J2] Saggu, S., **Daneshvar, H.**, Samavi, R., Pires, P., Sassi, R. B., Doyle, T. E., Zhao, J., Mauluddin, A., & Duncan, L. (2024). *Prediction of emergency department revisits among child and youth mental health outpatients using deep learning techniques*. BMC medical informatics and decision making, 24(1), 42. [\[Link\]](#)

[C1] **Daneshvar, H.**, & Samavi, R. (2022). *Heterogeneous Patient Graph Embedding in Readmission Prediction*. Proceedings of the Canadian Conference on Artificial Intelligence. [\[Link\]](#)

[J1] **Daneshvar, H.**, & Ravanmehr, R. (2022). *A social hybrid recommendation system using LSTM and CNN*. Concurrency and Computation: Practice and Experience, 34(18), e7015. [\[Link\]](#)

An updated list of my publications is available at my Google Scholar [\[Link\]](#)

PRESENTATIONS & ABSTRACTS

[PA12] Xie, S., Meyappan, V., Karimi, H., **Daneshvar, H.**, Kamaledin, MA., Samavi, R., Bhat, V., Sassi, R. (May 2026). *Evaluating the Open-source Toxicity Detector, "Detoxify", in Sensitive Migration and Health Contexts*. Trust to Impact: Trustworthy and Responsible AI for Citizenship, Migration, and Social Good (TIAI), Toronto, ON, Canada.

[PA11] Bonert, M., Ho, E., Wang, F., **Daneshvar, H.**, Naqvi, A. (Sep 2025). *Utilizing the report structure when extracting information from free text reports via natural language processing and an open large language model*. European Congress of Pathology, Vienna, Austria.

[PA10] **Daneshvar, H.** and Samavi, R. (Jul 2025). *GNN's Uncertainty Quantification using Self-Distillation*. Vector Institute ML Privacy and Security Workshop, Toronto, ON, Canada.

- [PA9] **Daneshvar, H.** and Samavi, R. (Mar 2025). *Uncertainty Quantification in Graph Neural Networks*. Vector Institute Research Symposium Poster Presentation (Remarkable 2025), Toronto, ON, Canada.
- [PA8] **Daneshvar, H.** (Dec 2024). *Trustworthy Graph Neural Networks*. McMaster University CSE Seminar, Hamilton, ON, Canada.
- [PA7] **Daneshvar, H.** and Samavi, R. (Jul 2024). *GCE: Confidence Calibration Error for Improved Trustworthiness of GNNs*. Vector Institute ML Security & Privacy Workshop, Toronto, ON, Canada.
- [PA6] **Daneshvar, H.**, Zhao, J., Mauluddin, A., Duncan, L., Pires, P., Sassi, R., Samavi, R., Doyle, T. (Jun 2024). *Graph Data Fusion to Predict Emergency Department Visit within 180-Days*. Precision Child and Youth Mental Health Conference, Ottawa, ON, Canada.
- [PA5] **Daneshvar, H.**, Saggu, S., Zhao, J., Mauluddin, A., Duncan, L., Pires, P., Sassi, R., Samavi, R., Doyle, T. (Jun 2024). *GNN in 30-Day ED Prediction for Child/Youth*. Precision Child and Youth Mental Health Conference, Ottawa, ON, Canada.
- [PA4] **Daneshvar, H.** and Samavi, R. (Feb 2024). *Confidence Calibration Loss for Graph Neural Networks*. Vector Institute Research Symposium Poster Presentation (Remarkable 2024), Toronto, ON, Canada.
- [PA3] **Daneshvar, H.**, Rashidiani, S., Zhao, J., Mauluddin, A., Boursalie, O., Duncan, L., Pires, P., Sassi, R., Samavi, R., Doyle, T. (Oct 2023). *Predicting Child and Youth Mental Health Service Use with Deep Learning Models*. Canadian Psychiatric Association Annual Conference, Vancouver, BC, Canada.
- [PA2] **Daneshvar, H.** and Samavi, R. (Feb 2023). *Questionnaire Graph Embedding for Early Prediction of Mental Health Emergency Department Admission*. Vector Institute Research Symposium Poster Presentation, Toronto, ON, Canada.
- [PA1] **Daneshvar, H.** and Samavi R. (Feb 2022). *Using Graph Neural Networks in Mental Health Service Utilization*. Vector Institute Research Symposium Poster Presentation, Toronto, ON, Canada.